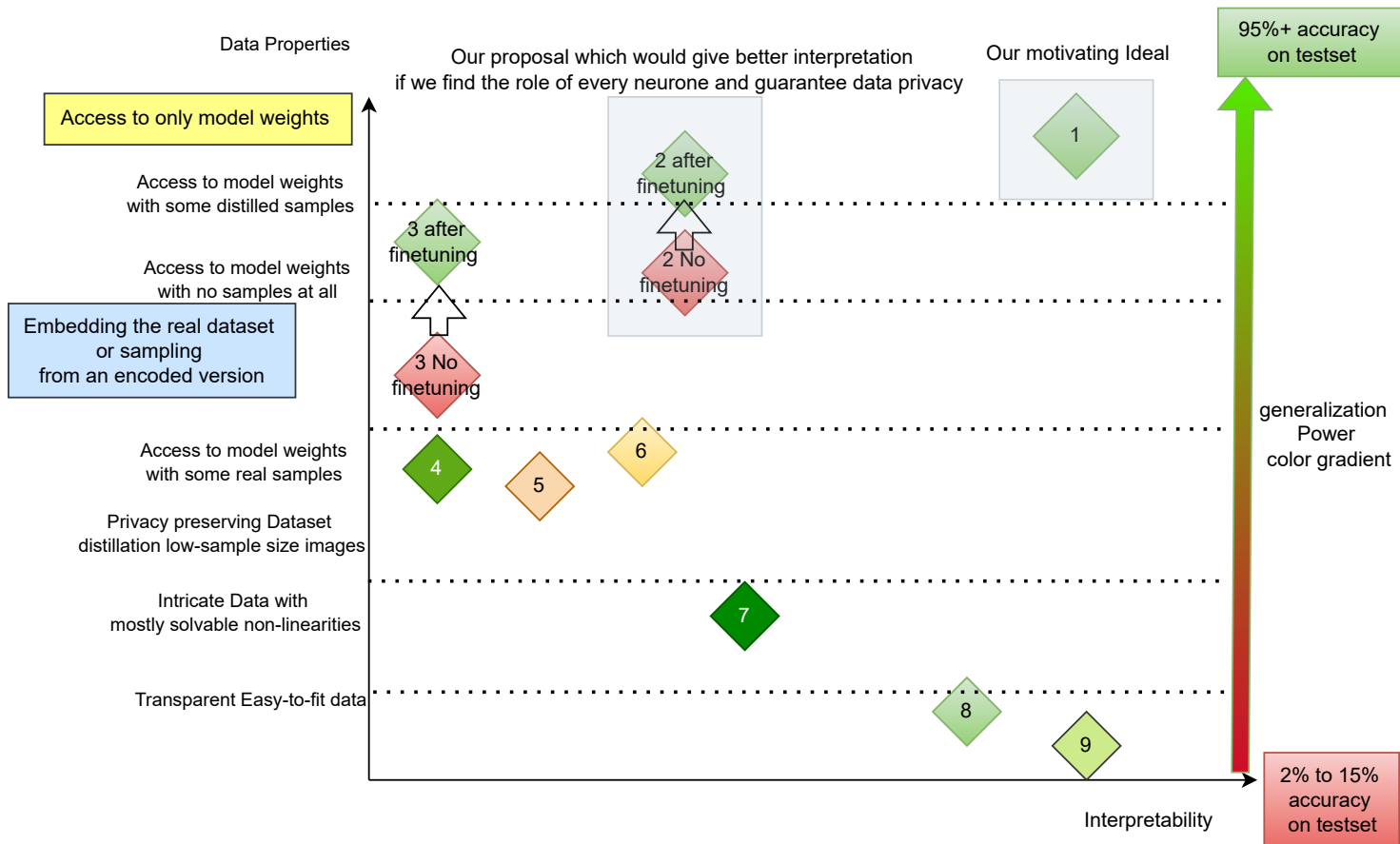




Model training

Accelerations

Hugging face accelerate

Parallelization

native pytorch DDP

(had access to a 3 Tesla T4 GPUs server)
3060/3090 RTX 1h7minutes/epoch
1Tesla T4 4h20minutes/epoch
parallelized 3Tesla T4 2h38minutes/epoch

Horovod

metrics

F1 score of predicted and finetuned CNNs

MSE

LWLN

MAE /MAPE

Bottleneck distance between
predicted weights and finetuned weights ?
more examples in "Loss function"

Architecture

when to do the fusion

<https://arxiv.org/pdf/1511.06434> showed we can do element wise
addition and substractions in the latent space can we make our AE
give such a latent space ?

replace 2 encoders with a single double sized one
Before/after each Encoder

after stacking the 2 encoder embeddings and before the dense Layer

after dense Layer and before decoder

before or after the decoder prediction normalization layer

Should we even be keeping the normalization layer ?
could we replace normalization with just kernel density estimation ?

Should we add class labels embeddings and if yes : where ?

almost same bullet points as "where to do fusion"

Should i use more advanced NLP techniques ?

Chain of Thought Longer Context/max token size

LLMs can be merged . on what basis is this effective ?

limit the cross-attention to individual layers ?

GCN , GNN , GATs for 1-simplices

Loss function

Should we somehow represent the gap between
the predicted model and finetuned version in the loss ?
(would we be cheating ?)

MSE

mahalanobis distance

LWLN

Extend Frechet Inception distance

Haversine distance

G-edit on mapper graphs

Fischer Information metric

Gromov-Wasserstein

distances on the weight space
(represented as a weight matrix's) Laplacien

KL-divergence

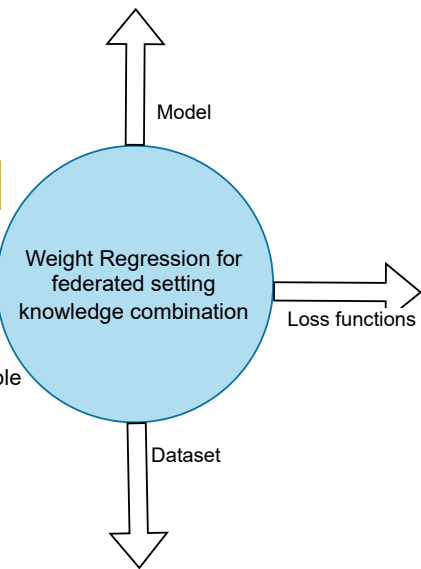
a convex representation of the desired weights norm as
higher normed ones/more similar to target model's
<https://arxiv.org/abs/1902.08160v4>

Geodesic distances <https://arxiv.org/pdf/2106.10777>

Spectral analysis and creating a locally convex distance between
eignvalues of finetuned weight matrices and predicted ones

Sinkhorn / Wasserstein

using scipy version does not tarrk the pytorch
computational graph.this worked because the loss was
propagated to MSE's computational graph .
Geomloss defines wasserstein better



Datasets

all combinations of MNIST

<https://github.com/ModelZoos/ModelZooDataset>

Try model zoos for non computer vision tasks:
community detection , time-series(TDA very helpful here)

Simulate Konstantin Schurholt
neuron-level data augmentation
leveraging symmetries in the weight space

Dataset Distillation mini-zoos

Created high-low performance models

use other metrics to decide this

Federated learning model similarities

Produce Adversarial learning-Proof Datasets

we need normal image real data in an average of 175 steps of 75 batch
size to conclude the first epoch of finetuning and regain strong performance .
what's the minimum number of images per class to do this fine tuning ?

the original DD paper explains that in theory the number of distilled images
of a dataset must be at least equal or above the number of classes .
Recent works have proved that you can combine multiple classes in
the same distilled images via more advanced distillation methods

can we do zoo-level distillation ?

Some commercial companies participate in federated learning (car manufacturing rivals for example)
Company A knows a member of the circle of clients that Company B participate in federated learning with
Company A launches an adversarial learning atack that changes a google board text completion/suggestion from "Company B cars are good" to "Company B cars are sub-par"

intro to fed learning <https://www.youtube.com/playlist?list=PLoEMreTa9CNmPnHffTFkPpbCJWpzm4In> during which a paper was brought up <https://arxiv.org/pdf/1807.00459>

Fig. 1 gives a high-level overview of this attack. Model replacement takes advantage of the observation that a participant in federated learning can (1) directly influence the weights of the joint model, and (2) train in any way that benefits the attack, e.g., arbitrarily modify the weights of its local model and/or incorporate the evasion of potential defenses into its loss function during training. We demonstrate the power of model replacement on two concrete learning tasks from the federated-learning literature: image classification on CIFAR-10 and word prediction on a Reddit corpus. Even a single-shot attack, where a single attacker is selected in a single round of training, causes the joint model to achieve 100% accuracy on the backdoor task. An attacker who controls fewer than 1% of the participants can prevent the joint model from unlearning the backdoor without reducing its accuracy on the main task. Model replacement greatly outperforms "traditional" data poisoning: in a

word-prediction task with 80,000 participants, compromising just 8 is enough to achieve 50% backdoor accuracy, as compared to 400 malicious participants needed for the data-poisoning attack.

models are a condensed version of data . distilled data is another distilled version of it, the gradients of training steps transport the manifold information.
can we quantify that information and use model-data similarities to do 1-step model weight regression ?